

Master PRD

LinkedIn Automation & AI Outreach Platform

Version 1.0 · Internal Build Document · Confidential

"We build the infrastructure. You own the tool. No SaaS dependency."

Product Name	Markeye Outreach Engine (codename: PHANTOM-X)
Version	1.0 — Initial Build
Author	Nihal Mishra — Founder, Markeye
Date	May 17, 2026
Status	APPROVED FOR BUILD — Phase 1 starts immediately
Target Launch	Phase 1 MVP: 7 days from today · Phase 2: Day 21 · Phase 3: Day 45
Revenue Target	\$10,000 MRR by Month 3 · \$20,000 MRR by Month 6
Stack	Python · FastAPI · Playwright · Supabase · n8n · Redis · Claude API · Docker

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01 Executive Summary & Product Vision

Problem	LinkedIn outreach tools cost \$69–\$439/mo, run on unsafe datacenter IPs causing account bans, offer zero AI personalization, and lock agencies into per-seat pricing. 40% of user reviews cite bans. 60% cite pricing frustration. Nobody has built: affordable + safe + AI-native + self-hostable + agency-ready.
Solution	Markeye Outreach Engine — a self-hosted LinkedIn automation platform with residential proxy routing built-in, Claude AI writing every message, multi-account rotation for agencies, and white-label deployment. One-command Docker setup. Flat pricing. Client owns their data.
Market	LinkedIn automation market: \$850M in 2026, growing 42% YoY. 80% of B2B leads come from LinkedIn. Every B2B company needs outreach. Generic messages now get 60% lower response rates — AI personalization is the new baseline.

Product Vision Statement

■ The only LinkedIn outreach tool that is safe by default, writes every message with AI, runs on your own infrastructure, and costs 80% less than competitors — while outperforming them.

Core Design Principles

- Safe by Default:** Residential proxies, warm-up mode, daily limits — configured out of the box. User never touches a safety setting.
- AI-First, Not AI-Add:** Claude API is baked into every outreach step. Not a credit add-on. Not optional. Every message is AI-written.
- You Own the Stack:** Self-hosted on client's VPS. Their data never touches our servers. One Docker Compose file.
- Flat Simple Pricing:** No execution hours. No phantom slots. No credit burnout. One price, unlimited campaigns.
- Agency-Native:** Multi-tenant from day 1. Isolated workspaces. White-label. Master dashboard. Built for resale.

Success Definition

Metric	Month 1	Month 3	Month 6
MRR	\$0 (one-time)	\$10,000+	\$20,000+
LinkedIn ban rate (clients)	<5%	<2%	<1%
Avg connection accept rate	20%+	30%+	35%+
Avg reply rate	5%+	10%+	15%+
Clients live	2–3	8–10	20+

Metric	Month 1	Month 3	Month 6
Time to setup (new client)	<4 hours	<2 hours	<30 min

02 Market Context & Opportunity

Market Size

LinkedIn automation tools market: **\$850M in 2026, growing 42% YoY**. This is not a niche. B2B outreach is not going away — it's accelerating. The problem is existing tools are architecturally broken for how LinkedIn works in 2026.

Why Now Is the Right Time

- ✓ LinkedIn tightened daily limits in March 2026: 50 invites/day max (down from 100)
- ✓ Generic automated messages have 60% lower response rates vs 2024 — AI personalization is the only path
- ✓ Expandi at \$99/account × multiple accounts = unsustainable for agencies → they need a better solution
- ✓ PhantomBuster raised prices end of 2025 without notice → 25%+ user churn window open
- LinkedIn's detection algorithms upgraded in 2026 — datacenter IPs fail faster than ever
- Self-hosted / privacy-first tools gaining adoption as teams fear vendor lock-in

Competitive Gap Map

Competitor	Price	Safe IPs?	AI Msgs?	Multi-acct?	Self-hosted?	White-label?
PhantomBuster	\$69–439	✗	~	✗	✗	✗
Expandi	\$99/acct	✓	✗	~	✗	✗
HeyReach	\$79–199	~	✗	✓	✗	✗
Waalaxy	\$43–112	~	✗	✗	✗	✗
Dux-Soup	\$15–99	✗	✗	✗	✗	✗
Dripify	\$59–99	~	✗	~	✗	✗
La Growth Mach.	\$80–240	~	✗	~	✗	✗
MARKEYE ✓	\$20–50*	✓	✓	✓	✓	✓

* Infra cost only. Service pricing on top of this.

03 Target Users & ICP

P1 — Agency Nihal

B2B Agency Owner (\$1M–\$10M ARR)

Context:

Runs a growth agency. Manages LinkedIn outreach for 5–15 clients. Currently paying \$79–\$199/mo per client on HeyReach or Expandi. That's \$400–\$3K/mo just in tools. Needs white-label, multi-account, and clean data isolation between clients.

Pain Points:

- Paying \$99/account × 8 clients = \$792/mo for Expandi alone
- Client data mixing — accidental cross-campaign sends
- No white-label — clients see the tool brand, not his agency
- LinkedIn bans one client account = angry email, refund demand
- Manually tracking campaign stats across 8 dashboards

Wants / Goals:

- One dashboard for all clients
- Each client in isolated workspace — zero data bleed
- White-label: client sees 'Powered by [Agency]', not Markeye
- Flat pricing — not per-account, not per-seat
- Auto-alerts when a campaign fails or hits a limit

Job-to-be-Done:

When managing multiple client LinkedIn campaigns, I want to run everything from one interface so that I can scale my agency without scaling my tool spend.

P2 — Founder Felix

AI-native SaaS Founder (\$500K–\$5M ARR)

Context:

Solo or small team founder. Doing his own outbound to find enterprise clients. Tried PhantomBuster, got banned. Tried Expandi, paying \$99/mo for one account. Knows Python. Wants a tool he can actually trust and tweak.

Pain Points:

- PhantomBuster banned his LinkedIn account — lost 2,000 connections
- Existing tools use datacenter IPs — LinkedIn flags them immediately
- No AI personalization — his messages look like everyone else's
- Reply rate stuck at 2–3% with template messages
- Doesn't want another SaaS subscription he can't control

Wants / Goals:

- Residential IP routing — campaigns that actually survive
- AI-written messages that reference prospect's actual LinkedIn content
- Self-hosted — full control, data stays on his server
- API access to integrate with his existing CRM/stack
- Reply tracking + CRM pipeline in one view

Job-to-be-Done:

When doing LinkedIn outreach at scale, I want AI to write personalized messages from scraped profile data so that my reply rates look like 1:1 outreach, not bulk spam.

P3 — SDR Sam

Sales Development Rep at \$5–50M ARR SaaS

Context:

SDR at a Series A/B SaaS company. Has a quota of 30 booked demos/month. Uses LinkedIn + email. Not technical — needs plug-and-play. Currently using Waalaxy or Dux-Soup but hitting ban warnings monthly.

Pain Points:

- Waalaxy keeps warning about connection limits — hits cap in 10 days
- Reply rate tanked in 2025 — same template going to same personas
- Has to keep browser open for Dux-Soup — laptop can't go to sleep
- CRM sync is broken — manually exporting CSVs to HubSpot weekly
- Manager wants reporting he can't generate from current tools

Wants / Goals:

- Cloud-based — runs 24/7 without laptop being on
- HubSpot/Salesforce native sync — zero manual exports
- AI messages that write themselves from prospect data
- Simple campaign builder — not a 20-step setup
- Dashboard with accept rate, reply rate, booked meetings

Job-to-be-Done:

When I set up a LinkedIn campaign, I want it to run automatically with AI-personalized messages so that I can hit my demo quota without spending 3 hours/day on manual outreach.

04 Product Architecture

System Architecture Overview

Markeye Outreach Engine is a **containerized, event-driven automation platform**. The frontend triggers campaigns via the dashboard or n8n. The API layer queues jobs. The Playwright engine pulls from the queue, executes via residential proxy, stores results in Supabase, and fires webhooks on completion.

Layer	Component	Technology	Role
Presentation	Dashboard UI	React + Tailwind	Campaign manager, lead CRM, analytics, reply inbox
Orchestration	Workflow Engine	n8n (self-hosted)	Campaign scheduling, webhook triggers, multi-step flows
API	REST API	FastAPI (Python)	Campaign CRUD, job dispatch, results retrieval, auth
Queue	Job Queue	BullMQ + Redis	Async Playwright job queue, retry logic, rate limiting
Automation	Browser Engine	Playwright + stealth	LinkedIn session simulation, profile scraping, actions
Proxy	IP Routing	Decodo/Brightdata	Residential IP per session, rotation on block
AI	Personalization	Claude API (Haiku)	Profile analysis, message writing, ICP scoring
Storage	Database	Supabase (Postgres)	Leads, campaigns, sessions, results, analytics
Cache/Session	Session Store	Upstash Redis	li_at cookies, session state, job queue, rate limits
Email Enrichment	Enrichment Pipeline	Apollo/Hunter waterfall	Email discovery from LinkedIn profile data
Deploy	Containerization	Docker Compose	One-command deploy: all services in one compose file
Proxy/HTTPS	Reverse Proxy	Caddy	Auto HTTPS, subdomain routing, auth middleware
Monitoring	Alerts	Slack webhooks	Campaign failures, ban detection, quota alerts

Data Flow: Campaign Execution

Step	Description
1. TRIGGER	User creates campaign in Dashboard OR n8n sends POST /campaign/run
2. QUEUE	FastAPI validates request → pushes job to BullMQ Redis queue with priority
3. PICK UP	Playwright worker picks job from queue → fetches li_at from Redis session store
4. PROXY	Worker requests residential IP from Decodo API → configures Playwright context
5. EXECUTE	Playwright navigates LinkedIn with session cookie → visits profile → scrapes data
6. AI WRITE	Profile data sent to Claude Haiku API → personalized connection message generated
7. ACTION	Playwright sends connection request with AI message → random delay (3–8s) applied
8. STORE	Result (accepted/pending/blocked) written to Supabase leads table
9. CHECK	Ban detection: if 401/429 → auto-pause campaign → Slack alert → queue suspended
10. WEBHOOK	On campaign batch complete → POST to n8n webhook → triggers follow-up flow

Docker Compose Service Map

Full stack in one compose file. Client runs **docker compose up** and everything starts.

Service	Image	Port	Depends On	Notes
api	python:3.12 + FastAPI	8000	db, redis, playwright	Main REST API. Exposes /campaigns /leads /accounts /webhooks
playwright	mcr.microsoft/playwright	—	redis, proxy	Headless Chromium workers. Scale: docker compose scale playwright=3
n8n	n8nio/n8n	5678	db, redis	Workflow orchestrator. Webhook triggers + scheduling
redis	redis:7-alpine	6379	—	Session store + BullMQ queue backend
caddy	caddy:2-alpine	80/443	api, dashboard, n8n	Reverse proxy + auto TLS. Routes subdomains
dashboard	node:20 (React build)	3000	api	Campaign UI. Built and served via Caddy
db (Supabase)	Supabase cloud	5432	—	Managed Postgres. Or self-host with supabase/postgres image

05 Feature Specifications (Epics & User Stories)

15 epics, each with user story and acceptance criteria. Priority = business value. Effort = engineering days.

EPIC 1 — Account & Session Management

US-01: Add LinkedIn Account

Priority: **HIGH** Effort: **LOW**

As an agency owner, I want to add multiple LinkedIn accounts so that I can rotate senders across campaigns.

Acceptance Criteria:

- User pastes li_at cookie OR uses browser extension to capture session
- Session stored encrypted in Redis with account_id key
- Session health check runs every 6 hours — 401 = flagged as expired
- Slack/email alert sent on session expiry
- Dashboard shows account status: Active / Expired / Warming Up / Banned

US-02: Account Warm-Up Mode

Priority: **HIGH** Effort: **LOW**

As a user with a new LinkedIn account, I want automatic warm-up so that I don't get banned on day one.

Acceptance Criteria:

- Warm-up schedule: Day 1–3: 5 actions/day, Day 4–7: 10/day, Day 8–14: 25/day, Day 15+: 50/day
- Actions during warm-up: profile visits, post likes, not connection requests
- Warm-up progress bar visible in account dashboard
- User can override and start campaigns early — with explicit warning shown
- Warm-up resets if account gets restricted

US-03: Multi-Account Campaign Rotation

Priority: **HIGH** Effort: **MED**

As an agency, I want campaigns to auto-rotate across N accounts so that I can hit higher daily volumes safely.

Acceptance Criteria:

- Campaign config: select accounts to include in rotation pool
- System auto-distributes leads across accounts (round-robin or weighted)
- Per-account daily cap enforced even in rotation
- If one account gets restricted, rotation continues with remaining accounts
- Rotation log visible per campaign: 'Account A: 45 sent, Account B: 42 sent'

EPIC 2 — Proxy & Safety Engine

US-04: Residential Proxy Routing

Priority: **HIGH** Effort: **MED**

As a user, I want every LinkedIn session to run through a residential IP so that LinkedIn can't detect I'm automating.

Acceptance Criteria:

- Each Playwright context launched with Decodo proxy credentials
- Proxy configured: country=target account's country, sticky session per account_id
- Proxy health check before every campaign run
- Auto-rotate IP if proxy returns 407 or LinkedIn returns 429
- Proxy cost displayed per campaign run in dashboard

US-05: Ban Detection & Auto-PausePriority: **HIGH** Effort: **MED**

As a user, I want campaigns to auto-pause when LinkedIn flags my account so that I don't make a ban worse.

Acceptance Criteria:

- Monitor for: HTTP 401, LinkedIn '/checkpoint' redirect, CAPTCHA page
- On detection: immediately pause all campaigns for that account_id
- Fire Slack webhook: 'Account [name] may be restricted. Campaigns paused.'
- Dashboard shows account status: 'Possible Restriction — Action Required'
- Campaign resumes only after user manually confirms account is healthy
- Soft block (warning): reduce daily limit by 50% for 7 days automatically

EPIC 3 — Lead Discovery & Scraping

US-06: LinkedIn Search ScraperPriority: **HIGH** Effort: **MED**

As a user, I want to scrape leads from a LinkedIn search URL so that I can build targeted prospect lists.

Acceptance Criteria:

- User pastes LinkedIn search URL or Sales Navigator search URL
- Playwright scrapes: name, headline, current role, company, location, profile URL, connection degree
- Pagination: auto-scroll through all result pages up to configured max (default 500)
- Deduplication: skip profiles already in leads table for this workspace
- Progress shown live: 'Scraped 127/500 profiles'
- Results stored in Supabase leads table immediately as scraped

US-07: AI-Powered Lead Discovery (No CSV Mode)Priority: **MED** Effort: **HIGH**

As a user, I want to describe my ICP in plain English and have the system find matching leads automatically.

Acceptance Criteria:

- User inputs: product description + target persona description (free text)
- Claude API generates 5–10 LinkedIn search queries from description
- System executes all searches and scrapes results
- Each scraped profile scored 0–100 for ICP fit by Claude
- Only profiles scoring 60+ added to campaign automatically
- User reviews scored leads with Accept/Reject before sending

US-08: Email Enrichment PipelinePriority: **MED** Effort: **MED**

As a user, I want to automatically find email addresses for my LinkedIn leads so that I can run multi-channel outreach.

Acceptance Criteria:

- For each lead, extract: company domain from profile
- Waterfall enrichment: try Apollo API → Hunter.io → Anymailfinder
- Store verified email in leads.email field
- Show enrichment confidence: Verified / Likely / Guessed / Not Found
- Email credits consumed displayed per campaign
- Failed enrichment: flag lead for manual review

EPIC 4 — AI Message Personalization

US-09: Claude-Powered Connection MessagePriority: **HIGH** Effort: **MED**

As a user, I want every connection request message to be uniquely written by AI for each prospect so that my reply rate exceeds industry average.

Acceptance Criteria:

- Before sending, scrape full profile: headline, about, recent posts (last 3), experience, skills
- Send profile data + campaign brief to Claude Haiku API
- System prompt: 'Write a <280 char LinkedIn connection message that references something specific from their profile. Sound human. No emojis. No generic openers.'
- Message preview shown in dashboard before campaign runs (sample 5 messages)
- User can add custom instructions: 'always mention our mutual connection in [city]'
- Fallback to template message if Claude API fails or profile data insufficient
- Message stored in leads table for CRM view

US-10: Follow-Up Message SequencePriority: **HIGH** Effort: **MED**

As a user, I want automated follow-up DMs after connection is accepted so that I can nurture leads without manual effort.

Acceptance Criteria:

- Sequence config: Step 1 (connect), Step 2 (DM after accept, delay configurable), Step 3 (follow-up if no reply)
- Each step: AI-written message based on lead data + sequence position context
- Delay between steps: configurable (default: 3 days, 7 days)
- Stop sequence on: reply received, lead marked 'Not Interested', manual override
- Sequence stats: connected / DM sent / replied / bounced per step

EPIC 5 — Campaign Manager

US-11: Create & Configure CampaignPriority: **HIGH** Effort: **MED**

As a user, I want a simple campaign wizard so that I go from zero to running campaign in under 5 minutes.

Acceptance Criteria:

- Step 1: Select/add LinkedIn accounts for campaign
- Step 2: Add leads (search URL OR CSV upload OR AI discovery)
- Step 3: Configure sequence (steps, delays, AI or template messages)
- Step 4: Set safety limits (max/day per account, run hours, max total)
- Step 5: Preview — see 3 sample AI messages before confirming
- Step 6: Launch — campaign starts, status shows in dashboard
- Campaign cloneable: 'Duplicate Campaign' with new lead list

US-12: Campaign Analytics DashboardPriority: **MED** Effort: **MED**

As a user, I want real-time campaign analytics so that I can optimize outreach without waiting for CSV exports.

Acceptance Criteria:

- Per campaign metrics: sent / accepted / acceptance rate / replied / reply rate / meetings booked
- Per account metrics: actions today / actions this week / safety score
- Time-series chart: daily accept rate trend
- Best-performing message variant (if A/B active) highlighted
- Export to CSV: one click, all lead data + outreach status
- Dashboard auto-refreshes every 60 seconds

EPIC 6 — CRM & Reply Inbox**US-13: Lead Pipeline CRM**Priority: **MED** Effort: **HIGH**

As a user, I want a Kanban-style lead pipeline so that I can track every lead from 'sent' to 'booked meeting'.

Acceptance Criteria:

- Pipeline stages: Queued → Sent → Connected → Replied → Interested → Meeting Booked → Closed
- Drag leads between stages manually
- Auto-move: Connected when LI accepts, Replied when DM reply detected
- Lead detail panel: full profile data, all messages sent, reply content, notes field
- Filter/sort by: campaign, stage, date, company, accept date
- Bulk actions: move to stage, assign to campaign, export

US-14: Unified Reply InboxPriority: **MED** Effort: **HIGH**

As a user, I want all LinkedIn replies across all accounts in one inbox so that I don't miss a hot lead.

Acceptance Criteria:

- Poll LinkedIn DM inbox for each connected account every 30 minutes
- New replies appear in unified inbox with: sender name, message preview, account used
- Reply from inbox: type response → system sends via correct LinkedIn account
- Mark as: Replied / Not Interested / Meeting Booked
- Auto-stop sequence when reply detected
- Alert: Slack/email notification on new reply

EPIC 7 — Agency & Multi-Tenant

US-15: Multi-Tenant Workspaces

Priority: **HIGH** Effort: **HIGH**

As an agency owner, I want each client to have an isolated workspace so that there is zero data bleed between clients.

Acceptance Criteria:

- Each client = separate workspace with own: leads, campaigns, accounts, API keys
- Agency master account can switch between client workspaces
- Data isolation enforced at DB level: workspace_id on every row with RLS policies
- Client can optionally be given their own login to view their workspace
- Workspace billing tracked separately for agency cost allocation

US-16: White-Label Configuration

Priority: **MED** Effort: **MED**

As an agency, I want to white-label the dashboard with my brand so that clients see my agency, not 'Markeye'.

Acceptance Criteria:

- Admin config: custom logo, primary color, product name, support email
- Dashboard header shows agency brand
- Email notifications sent from agency email (via SMTP config)
- Optional: custom domain (agency sets DNS CNAME to our Caddy reverse proxy)
- Client-facing URLs: app.agencyname.com instead of app.markeye.space

06 API & Integration Specification

REST API Endpoints

Method	Endpoint	Auth	Description
POST	/auth/login	None	Username+password → JWT token
POST	/auth/refresh	JWT	Refresh expired JWT
GET	/accounts	JWT	List all LinkedIn accounts in workspace
POST	/accounts	JWT	Add account (li_at cookie + label)
DELETE	/accounts/{id}	JWT	Remove account
GET	/accounts/{id}/health	JWT	Check session validity + safety score
GET	/campaigns	JWT	List campaigns (paginated)
POST	/campaigns	JWT	Create campaign (full config)
GET	/campaigns/{id}	JWT	Campaign detail + live stats
POST	/campaigns/{id}/start	JWT	Start/resume campaign
POST	/campaigns/{id}/pause	JWT	Pause campaign
DELETE	/campaigns/{id}	JWT	Delete campaign + leads
GET	/campaigns/{id}/leads	JWT	Leads for campaign (filterable)
GET	/leads	JWT	All leads in workspace (filterable)
PATCH	/leads/{id}	JWT	Update lead status/notes
POST	/leads/import	JWT	Bulk import leads from CSV
POST	/scrape/search	JWT	Trigger LinkedIn search scrape
POST	/scrape/profile	JWT	Scrape single LinkedIn profile
POST	/ai/score-leads	JWT	Score lead list against ICP description
POST	/ai/preview-message	JWT	Generate sample AI message for a lead
GET	/inbox	JWT	Unified reply inbox (all accounts)
POST	/inbox/{thread_id}/reply	JWT	Send DM reply via correct account
GET	/analytics/campaign/{id}	JWT	Campaign metrics time-series
GET	/analytics/workspace	JWT	Workspace-level aggregate stats
POST	/webhooks	JWT	Register webhook URL for events

Method	Endpoint	Auth	Description
GET	/webhooks	JWT	List registered webhooks

Webhook Events

All webhooks POST JSON to registered URL. Include workspace_id, event_type, timestamp, payload.

Event	Trigger	Payload includes
campaign.started	Campaign begins execution	campaign_id, account_ids, lead_count
campaign.paused	Campaign paused (manual or auto)	campaign_id, reason, leads_remaining
campaign.completed	All leads processed	campaign_id, stats summary
lead.connected	Connection request accepted	lead_id, lead_name, campaign_id
lead.replied	DM reply received	lead_id, message_preview, thread_id
account.restricted	Ban detection triggered	account_id, detection_type
account.session_expired	li_at cookie expired	account_id, label
enrichment.completed	Email enrichment batch done	campaign_id, found_count, failed_count

n8n Integration

The platform is designed as an n8n-native backend. Pre-built n8n workflow templates provided:

Template Name	Trigger	What It Does
LinkedIn Prospector	Schedule (daily 9am)	Scrape search → score → start campaign → notify Slack
Reply Handler	Webhook (lead.replied)	New reply → classify intent → update CRM → alert SDR
Multi-Step Sequence	Webhook (lead.connected)	Connect → wait 3d → send DM → wait 7d → follow-up
Ban Alert	Webhook (account.restricted)	Account banned → pause all → Slack alert → create task
Weekly Report	Schedule (Monday 8am)	Pull analytics → format → email to client/agency
CSV-to-Campaign	Manual trigger	Upload CSV → import leads → create campaign → start
Apollo Enrichment Chain	Webhook (scrape.done)	New leads → enrich emails → score ICP → add to sequence

07 Database Schema (Supabase / PostgreSQL)

Table: workspaces

Column	Type	Notes
id	uuid	PK, default gen_random_uuid()
name	text	Agency or company name
owner_id	uuid	FK → users.id
branding	jsonb	logo_url, primary_color, product_name
plan	text	'starter' 'pro' 'agency'
created_at	timestamptz	default now()

Table: users

Column	Type	Notes
id	uuid	PK
workspace_id	uuid	FK → workspaces.id, RLS enforced
email	text	Unique
role	text	'owner' 'admin' 'member' 'client_view'
created_at	timestamptz	default now()

Table: linkedin_accounts

Column	Type	Notes
id	uuid	PK
workspace_id	uuid	FK → workspaces.id
label	text	Human name: 'Nihal - Main'
li_at_encrypted	text	AES-256 encrypted session cookie
status	text	'active' 'warming_up' 'restricted' 'expired'
warmup_day	int	Current day in warm-up schedule (0=complete)
daily_limit	int	Max actions per day, default 50
actions_today	int	Reset daily via cron
proxy_country	text	ISO country code for proxy routing
last_health_check	timestamptz	Last session validation

Column	Type	Notes
created_at	timestampz	default now()

Table: campaigns

Column	Type	Notes
id	uuid	PK
workspace_id	uuid	FK → workspaces.id
name	text	Campaign display name
status	text	'draft' 'active' 'paused' 'completed' 'error'
account_ids	uuid[]	Array of linkedin_account IDs in rotation
sequence	jsonb	[[step, type, delay_days, message_template, use_ai]]
daily_limit	int	Max sends per day across all accounts
icp_description	text	Plain English ICP for AI scoring
stats	jsonb	{sent, connected, replied, meetings, accept_rate, reply_rate}
created_at	timestampz	default now()
started_at	timestampz	null until first run

Table: leads

Column	Type	Notes
id	uuid	PK
workspace_id	uuid	FK, RLS
campaign_id	uuid	FK → campaigns.id, nullable
profile_url	text	LinkedIn URL, unique per workspace
full_name	text	
headline	text	
current_title	text	
company	text	
location	text	
about	text	Scraped 'about' section
recent_posts	jsonb	Last 3 post texts for AI context
email	text	Enriched email (nullable)

Column	Type	Notes
email_confidence	text	'verified' 'likely' 'guessed' null
icp_score	int	0–100 AI-generated ICP fit score
pipeline_stage	text	'queued' 'sent' 'connected' 'replied' 'interested' 'booked' 'closed' 'not_interested'
sent_at	timestamptz	null until request sent
connected_at	timestamptz	null until accepted
replied_at	timestamptz	null until reply received
message_sent	text	Actual AI-generated message sent
notes	text	Manual notes field
account_id	uuid	Which LI account handled this lead
created_at	timestamptz	default now()

Table: messages

Column	Type	Notes
id	uuid	PK
lead_id	uuid	FK → leads.id
direction	text	'outbound' 'inbound'
content	text	Message text
sent_via	uuid	FK → linkedin_accounts.id
created_at	timestamptz	default now()

Table: jobs

Column	Type	Notes
id	uuid	PK
type	text	'scrape' 'connect' 'message' 'enrich' 'health_check'
status	text	'queued' 'running' 'done' 'failed' 'retrying'
payload	jsonb	Job-specific config
result	jsonb	Output data
retries	int	default 0, max 3
created_at	timestamptz	default now()
completed_at	timestamptz	null until done

■ Row Level Security (RLS) is enabled on ALL tables. Every query auto-filters by `workspace_id` from JWT claim. No client can ever read another client's data.

08 Security & Compliance

Authentication & Authorization

- ✓ JWT-based auth. 24-hour access tokens, 30-day refresh tokens
- ✓ Supabase RLS: every row filtered by workspace_id from JWT claims
- ✓ API keys: per-workspace keys for n8n/external integration (stored hashed)
- ✓ li_at cookies encrypted AES-256 at rest in Supabase — never logged
- ✓ All env secrets in Docker secrets or .env (never committed to git)

Rate Limiting & Safety Guardrails

Guardrail	Default Value	Configurable?	Enforcement
Max connection requests/day/account	50	Yes (10–100)	Hard stop in job queue
Min delay between actions	3–8 seconds (random)	No	Playwright wait before each action
Max profile visits/hour	20	Yes	Redis rate limiter
Warm-up days before campaigns	14	Yes	Account status check before job dispatch
Retry limit per job	3 attempts	No	BullMQ retry with exponential backoff
Auto-pause on 401/429	Immediate	No	Playwright response interceptor
Session refresh reminder	72 hours	No	Cron → Slack alert
Max concurrent Playwright workers	3	Yes	BullMQ concurrency setting

Data Privacy

All scraped data is publicly visible LinkedIn profile information. However:

- Do NOT store sensitive personal data beyond what's visible on LinkedIn public profiles
- Do NOT store li_at cookies in plaintext — AES-256 encrypt before DB write
- GDPR: provide 'Delete all data for workspace' function — purges all leads + logs
- LinkedIn ToS: users accept ToS risk explicitly during onboarding (consent checkbox)
- India IT Act 2000: no storage of Aadhaar, financial data, or biometrics

LinkedIn ToS Disclosure

■ **IMPORTANT:** LinkedIn automation violates LinkedIn's ToS. Users must be informed and consent. Include: (1) Onboarding consent checkbox, (2) Risk disclosure in docs, (3) Contract clause transferring ToS risk to client. This is the same risk ALL competitors expose users to.

09 Build Phases & Timeline

Overview

4 phases. Phase 1 = working internal tool. Phase 2 = first paying client. Phase 3 = productized agency tool. Phase 4 = self-serve SaaS.

Phase 1: Core Engine MVP

Days 1–7

Day	Task
Day 1	Fork OpenOutreach → strip DjangoCRM → migrate to FastAPI skeleton
Day 1	Supabase project setup: create all tables per schema in Section 07
Day 1	Upstash Redis: provision, connect, test BullMQ queue
Day 2	Playwright + stealth config: launch LinkedIn, verify no WebDriver flag
Day 2	Decodo proxy integration: route Playwright through residential IP
Day 2	Test: visit 20 LinkedIn profiles → confirm no detection, data stored
Day 3	Session management: store/retrieve li_at, health check endpoint
Day 3	Warm-up mode: implement day-based action scheduler
Day 4	Connection request flow: scrape profile → Claude message → send request
Day 4	Daily limit enforcer: Redis counter, hard stop at configured max
Day 5	Ban detection: intercept 401/429 → pause campaign → Slack alert
Day 5	n8n webhook trigger: POST /campaigns/{id}/start → runs campaign
Day 6	First real campaign: 20 connections to ICP targets with AI messages
Day 7	Bug fixes, stability testing, document setup process
Day 7	MILESTONE: Working automation tool. Use internally for Markeye outreach.

Phase 2: Client-Ready Version

Days 8–21

Day	Task
Day 8–9	Multi-account rotation: add N accounts, round-robin dispatch
Day 10	Email enrichment: Apollo API → Hunter waterfall → store results
Day 11	Sequence builder: multi-step campaigns (connect → DM → follow-up)
Day 12	Basic React dashboard: campaign list, lead table, status pipeline

Day 13–14	Campaign analytics: accept rate, reply rate, time-series charts
Day 15	Reply inbox: poll LinkedIn DMs → unified inbox → reply from UI
Day 16	Caddy setup: HTTPS, subdomain routing (api.*, app.*, n8n.*)
Day 17–18	Docker Compose: full stack, one-command deploy, tested on Hetzner
Day 19	Client onboarding docs: README + Loom walkthrough video
Day 20	First client deployment: install on client VPS, configure, train
Day 21	MILESTONE: First paying client live. Invoice sent. Case study begun.

Phase 3: Agency Platform

Days 22–45

Day	Task
Day 22–24	Multi-tenant workspaces: RLS policies, workspace switching
Day 25–26	White-label: logo, colors, product name, custom domain config
Day 27–29	AI lead discovery: ICP description → Claude generates queries → auto-score
Day 30–31	CRM pipeline: Kanban board, drag-drop, lead detail panel
Day 32–34	A/B message testing: two AI prompt variants, winner detection
Day 35–36	Safety score dashboard: per-account health metrics, risk indicators
Day 37–38	n8n workflow templates: 7 pre-built templates ready to import
Day 39–41	Agency master dashboard: all client workspaces in one view
Day 42–44	Billing integration (Stripe/Razorpay): subscription management
Day 45	MILESTONE: Agency-ready. Launch to 3 agency clients. Start \$10K MRR push.

Phase 4: Self-Serve SaaS

Days 46–90

Day	Task
Day 46–50	Self-serve signup: email → workspace created → guided onboarding
Day 51–55	Visual workflow builder: drag-drop campaign steps (n8n-style UI)
Day 56–60	Behavior triggers: 'send DM when target posts on LinkedIn'
Day 61–65	LinkedIn + Email waterfall: single campaign across both channels
Day 66–70	AI selector auto-recovery: detect DOM changes → Claude repairs selectors
Day 71–75	API marketplace: publish endpoints for Zapier, Make, Clay integration
Day 76–80	Analytics: cohort analysis, campaign benchmarks, A/B significance
Day 81–85	Mobile-responsive dashboard + Slack bot for quick campaign control

Day 86-90

MILESTONE: Self-serve live. \$20K MRR target. Product launches publicly.

10 Revenue Model & Pricing

Pricing Tiers

Tier	Price	Target	Includes	Limit
Starter	\$49/mo	Solo founders, SDRs	1 LI account, AI messages, basic CRM, n8n integration	500 actions/mo
Pro	\$99/mo	Small teams	5 LI accounts, multi-step sequences, email enrichment, reply inbox	2,000 actions/mo
Agency	\$299/mo	Agencies	Unlimited accounts, white-label, multi-tenant, priority support	10,000 actions/mo
White-Label	\$4,500 one-time	Clients of Markeye	Full stack on client VPS, 30-day support, source code access	Unlimited
Managed DFY	\$1,500–2,500/mo	Busy founders	We run campaigns for client, weekly reporting, meetings guaranteed	Per contract

90-Day Revenue Path

Month	Activities	One-Time Rev	MRR Added	Cumulative MRR
Month 1	2× white-label installs (\$4,500 each) + 1 consulting project (\$5K)	\$14,000	\$0	\$0 MRR
Month 2	3× Managed DFY retainers (\$2,000/mo) + 5× Pro seats (\$99/mo)	\$0	\$6,495	\$6,495 MRR
Month 3	+3 retainers + 10 Pro seats + 2 Agency plans (\$299/mo)	\$0	\$7,093	\$13,588 MRR
Month 4–6	Systemize: 10 retainer + 40 Pro + 6 Agency + SaaS self-serve growing	ongoing	\$20,000+	\$20,000+ MRR

■ Conservative scenario at 50%: \$6,800 MRR by Month 3. Still above break-even. At 100% you hit \$13.5K. Target is \$10K — that's between 50% and 75% of projections.

11 Success Metrics & KPIs

Product Health KPIs

KPI	Target (Month 1)	Target (Month 3)	How Measured
LinkedIn accept rate	20%+	30%+	connected/sent per campaign
LinkedIn reply rate	5%+	10%+	replied/connected per campaign
Account ban rate	<5%	<2%	restricted accounts / total accounts
Session expiry rate	<10%/week	<5%/week	expired sessions / total sessions
Campaign success rate	90%+	95%+	campaigns completed without error
AI message quality score	>7/10	>8/10	User ratings on sampled messages
Time to first campaign (new)	<4 hours	<2 hours	Onboarding → first send timestamp
Proxy detection rate	<5%	<2%	429/captcha per 1000 requests

Business KPIs

KPI	Month 1	Month 3	Month 6
MRR	\$0	\$10K+	\$20K+
Active clients	2–3	8–10	25+
Churn rate	—	<5%/mo	<3%/mo
NPS	—	>50	>60
Support tickets/wk	<5	<10	<15
Avg deal size	\$4,500	\$2,500	\$1,800

12 Risk Register

Risk	Severity	Probability	Mitigation	Owner
LinkedIn bans client account	CRITICAL	HIGH	Residential proxies + warm-up + <50 actions/day + auto-pause on warning	Nihal
LinkedIn detects Playwright	HIGH	MED	puppeteer-extra-stealth + fingerprint rotation + rebrowser-patches + real profile warm-up	Nihal
Proxy IP blacklisted	MED	MED	Sticky session only, rotate IP on 429, Decodo has 115M+ IPs — rotation pool is huge	Nihal
li_at session expires mid-campaign	MED	HIGH	72-hour expiry check cron → Slack alert → auto-pause → campaign resumes on refresh	Nihal
LinkedIn DOM changes break scrapers	MED	MED	Phase 4: AI selector recovery. Near-term: monitor breakage, fix within 24 hours	Nihal
Claude API rate limits or outage	MED	LOW	Fallback to template message on API error. Message still sends, just not personalized	Nihal
Client scales too fast, gets banned	HIGH	MED	Hard daily cap enforced in code. Client cannot override without code change	Nihal
Decodo proxy service outage	HIGH	LOW	Secondary proxy in config (Brightdata). Auto-failover logic in proxy middleware	Nihal
Data breach (li_at cookies stolen)	CRITICAL	LOW	AES-256 encryption at rest, TLS in transit, no logs of session values, Redis auth	Nihal
Legal: LinkedIn sues	CRITICAL	LOW	ToS disclosure + client consent. Same risk Expandi/\$99M HeyReach carry. Legal indemnity in contracts	Legal
India IT Act compliance	MED	LOW	No PII beyond public LI data. No financial data. Privacy policy + DPA with clients	Legal
Competitor clones feature	LOW	HIGH	Speed of iteration + agency relationships = moat. Ship faster than they copy	Product

13 Open Source Dependencies & Licenses

Repo / Package	Language	License	Stars	Role in Product
eracle/OpenOutreach	Python	MIT	Active	PRIMARY FORK — base LinkedIn automation engine
microsoft/playwright	Py/JS	Apache 2	70k+	Core browser automation. Use Python version.
berstend/puppeteer-extra	Node.js	MIT	7k+	Stealth plugin: disable WebDriver, canvas, CDP flags
rebrowser/rebrowser-patches	Node.js	MIT	1k+	2026 stealth patches — more current than above
browser-use/browser-use	Python	MIT	40k+	AI agent browser control for complex multi-step flows
joshiayush/inb	Python	Apache 2	600+	LinkedIn Voyager API client — more stable than DOM
apify/crawlee-python	Python	Apache 2	5k+	Robust crawler framework with built-in proxy rotation
n8nio/n8n	Node.js	FSL-1.1	47k+	Workflow orchestration — self-hosted OK under FSL
supabase/supabase	TS/Py	Apache 2	75k+	Database + auth + realtime + storage
filip-michalsky/SalesGPT	Python	MIT	2.2k	Reference: AI sales agent architecture patterns
taskforcesh/bullmq	Node.js	MIT	6k+	Job queue for async Playwright execution
pydantic/FastAPI	Python	MIT	80k+	REST API layer
caddyserver/caddy	Go	Apache 2	60k+	Reverse proxy + auto HTTPS
upstash/upstash-redis	TS/Py	MIT	3k+	Serverless Redis client for session + queue

■ **IMPORTANT: n8n uses FSL-1.1 (Functional Source License). Self-hosting for internal/client use is OK. Reselling n8n itself as a product is NOT allowed. We use n8n as an orchestrator — this is permitted.**

MASTER PRD v1.0 — Approved for Build — Phase 1 Starts Today

Target: \$10,000 MRR by Month 3 · \$20,000 MRR by Month 6

First action: git clone <https://github.com/eracle/OpenOutreach>

Markeye · Master PRD v1.0 · May 17, 2026 · Confidential · Author: Nihal Mishra